
Defying Gravity: The Biomechanics of Snake Locomotion

[0:00 - 1:15] Introduction & Motivation (開場與在地觀察動機)

(舞台指示：站在海報左側 *Motivation* 區塊旁，語氣充滿熱情，展現從生活環境中發掘問題的科學精神。)

"Good morning, respected judges and fellow researchers. Today, I am thrilled to present our biomechanical research titled: 'Biomechanical Analysis of Locomotion and Morphological Adaptations in Snakes from Distinct Habitats.'

Our research was inspired by our own school campus in Taiwan. Because we are located near mountainous areas, we frequently encounter the *Boiga kraepelini*, an arboreal cat snake. We observed its superior climbing and gap-crossing abilities and were fascinated. Without limbs, how do they overcome gravity?

This sparked our core research questions: What specific anatomical structures grant arboreal snakes an advantage in climbing? And how do their biomechanical strategies differ from terrestrial snakes?

[1:15 - 2:45] Experimental Subjects & Morphological Data (實驗對象與形態數據)

(舞台指示：手指向 *Experimental Animals* 區塊，隨後移動至 *Result 1* 的 *Body dimensions* 圖表。)

"To investigate this, we selected four snake species representing different habitats:

Snake A is a semi-arboreal Corn snake.

Snake B is a terrestrial Western Hognose.

Snake C is our local arboreal expert, the Cat snake.

And Snake D is a massive semi-arboreal *Boa constrictor*.

We designed an experimental crossing platform to measure their vertical extension and horizontal reaching limits. But before looking at their movement, we must look at their anatomy.

Please direct your attention to the 'Body dimensions' chart. We divided the snakes into Head, Trunk, and Tail. Look closely at Snake C, the arboreal specialist. It possesses a significantly larger proportion of tail length compared to the terrestrial Snake B. In biomechanics, this long tail acts as a crucial posterior anchor and a counterweight, allowing the snake to extend its anterior body forward without falling.

[2:45 - 4:15] Data Interpretation: Biomechanical Limits (數據解讀：生物力學的極限)

(舞台指示：走到海報中間，指向 *Vertical extension* 與 *Horizontal reaching* 的圖表，語氣轉為客觀嚴謹。)

"Now, let us examine their actual performance.

Looking at the 'Vertical extension' chart, Snake C unsurprisingly dominates, utilizing its long tail anchor to reach upward effectively.

However, the 'Horizontal reaching' data revealed a fascinating universal boundary. When we tested the maximum horizontal reach across different snake species, we discovered that despite their different habitats, their horizontal reach converges at a biological limit of **36% to 43% of their total body length**.

Why is there a limit? We conducted a mechanical analysis. Extending horizontally creates a massive downward anterior torque, especially if the snake has a heavy head. Once the extended body length exceeds roughly 40%, the anterior torque completely overpowers the posterior grip, and the snake falls. This is a fundamental physical limit governed by gravity.

[4:15 - 5:45] The Exception: Muscular Compensation (反直覺發現：肌肉代償機制)

(舞台指示：雙手微微張開，將重點轉向 *Snake D (Boa constrictor)*，展現科學研究中發現「例外」的驚喜感。)

"But biology is full of beautiful exceptions. Let's look at Snake D, the *Boa constrictor*. Morphologically, the Boa is at a severe disadvantage for crossing gaps. It has a massive, heavy body and a relatively short tail. According to our morphological trade-off theory, it should perform terribly because it lacks a long counterweight.

Yet, looking at our data, Snake D performs exceptionally well in vertical extensions and reaching. How is this possible?

Through our analysis, we concluded that this reveals a different locomotor strategy: **Muscular Compensation**. The *Boa constrictor* possesses vastly superior muscular tension. It uses sheer core strength to counteract the anterior torque, proving that extreme muscular strength can successfully compensate for morphological disadvantages.

[5:45 - 7:00] Conclusion & SDG Impact (結論與永續發展目標)

(舞台指示：退後一步，面向所有評審，將重點帶向 *SDGs* 與生態保育的實際應用。)

"In conclusion, our research successfully decodes the trade-offs in snake locomotion. A long tail provides a biomechanical anchor, while a heavy head limits horizontal reach to around 40% of body length, unless overcome by superior muscular tension.

Beyond physics, this research has practical ecological value, directly supporting the United Nations' SDG 15: Life on Land. By scientifically understanding the precise climbing limits and reaching capabilities of different snake species, we can design better, scientifically-informed anti-snake barriers for human settlements. This allows us to protect human residences without harming the local wildlife, fostering true 'Scientifically Informed Coexistence'.

Thank you for your time and attention. I am now happy to answer your questions."

 報告小叮嚀 (Tips for Delivery) :

1. **強調「極限數值 (36%-43%)」**：當您唸到 "converges at a biological limit of 36% to 43%" 時，請務必放慢語速並加重語氣。這展現了您將「生物學」轉化為「精確物理數學」的科學成熟度。
2. **對比的戲劇張力**：在講述蟒蛇 (*Boa constrictor* / *Snake D*) 那段時，先強調牠的「先天劣勢 (short tail, massive body)」，再點出牠用「肌肉力量 (muscular tension)」克服了物理限制。這種有轉折的科學推論會讓評審聽得津津有味。

這是一份專為您的「蛇類移動與形態適應的生體力學分析」海報量身打造的 **15 題中英文問答速查表 (Q&A Cheat Sheet)**。這份清單涵蓋了研究動機、形態學特徵（尾巴與頭部的力學作用）、反直覺的「水平極限」與「肌肉代償」，以及生態保育的實際應用。

 Q&A 速查表：克服重力的蛇類生物力學

1. 研究動機 (Motivation)

- **EN:** What inspired you to research the biomechanics of snake locomotion?
- **CH:** 是什麼啟發您研究蛇類移動的生物力學？
- **Answer (EN):** Our school campus is near the mountains, and we frequently observe the arboreal cat snake (*Boiga kraepelini*) demonstrating superior climbing and gap-crossing abilities.
- **Answer (CH):** 我們的校園靠近山區，經常觀察到樹棲型的「大頭蛇」展現出卓越的攀爬與穿越間隙能力。

2. 核心問題 (Core Question)

- **EN:** What is the main biological question you wanted to answer?
- **CH:** 您想解答的主要生物學問題是什麼？
- **Answer (EN):** We wanted to know what specific anatomical structures grant arboreal snakes an advantage in climbing and overcoming gravity compared to terrestrial snakes.
- **Answer (CH):** 我們想知道與陸棲蛇類相比，是哪些特定的解剖結構賦予了樹棲蛇類攀爬與克服重力的優勢。

3. 實驗對象 (Experimental Subjects)

- **EN:** Why did you choose these four specific snake species for your experiment?
- **CH:** 為什麼您選擇這四種特定的蛇類進行實驗？
- **Answer (EN):** We selected them to represent distinct habitats: arboreal, semi-arboreal, and terrestrial, to compare their morphological and behavioral differences.
- **Answer (CH):** 我們選擇牠們來代表不同的棲息環境：樹棲、半樹棲與陸棲，以比較牠們在形態與行為上的差異。

4. 身體比例劃分 (Body Dimensions)

- **EN:** How did you divide the snake's body to measure its morphological dimensions?
- **CH:** 您是如何劃分蛇的身體來測量其形態尺寸的？
- **Answer (EN):** We divided their body length into three specific parts: the Head, the Trunk, and the Tail (from vent to the tail tip).
- **Answer (CH):** 我們將牠們的體長分為三個特定部分：頭部、軀幹和尾部（從泄殖腔到尾尖）。

5. 樹棲蛇的形態優勢 (Arboreal Morphological Advantage)

- **EN:** According to your data, what morphological feature distinguishes the arboreal snake (Snake C)?
- **CH:** 根據您的數據，樹棲蛇（蛇 C）在形態上有什麼突出的特徵？
- **Answer (EN):** Snake C possesses a significantly larger proportion of tail length compared to terrestrial snakes.
- **Answer (CH):** 與陸棲蛇類相比，蛇 C（大頭蛇）擁有比例顯著較長的尾部。

6. 尾巴的力學功能 (Biomechanical Function of the Tail)

- **EN:** What is the biomechanical function of a long tail when a snake is crossing a gap?
- **CH:** 當蛇在穿越間隙時，長尾巴的生物力學功能是什麼？
- **Answer (EN):** The long tail acts as a crucial posterior anchor and a counterweight, preventing the snake from falling forward.
- **Answer (CH):** 長尾巴扮演著關鍵的「後方錨點（固定點）」與「配重塊」的角色，能防止蛇向前墜落。

7. 垂直延伸表現 (Vertical Extension Performance)

- **EN:** Which snake performed the best in vertical extension and why?
- **CH:** 哪一種蛇在垂直延伸表現最好？為什麼？

- **Answer (EN):** Snake C performed the best because it utilizes its long tail anchor effectively to reach upward.
- **Answer (CH):** 蛇 C 表現最好，因為牠能有效地利用長尾巴作為錨點來向上延伸。

8. 水平延伸極限 (Horizontal Reaching Limit)

- **EN:** What was the most surprising discovery in your horizontal reaching experiment?
- **CH:** 在水平延伸實驗中，您最驚訝的發現是什麼？
- **Answer (EN):** We found a biological limit: regardless of habitat, all snakes' horizontal reaching capability converges at around 36% to 43% of their total body length.
- **Answer (CH):** 我們發現了一個生物學極限：無論棲息地為何，所有蛇類的水平延伸能力都收斂在總體長的 36% 到 43% 左右。

9. 力學極限原因 (Cause of the Mechanical Limit)

- **EN:** Why is there a horizontal limit of around 40% for the snakes?
- **CH:** 為什麼蛇類會有大約 40% 的水平延伸極限？
- **Answer (EN):** Extending horizontally creates a massive downward anterior torque. Beyond 40%, gravity overpowers their posterior grip.
- **Answer (CH):** 水平延伸會產生巨大的向下「前段扭矩 (Anterior torque)」，超過 40% 時，重力會壓過牠們後方的抓握力。

10. 頭部重量的影響 (Impact of a Heavy Head)

- **EN:** How does having a heavy head affect a snake's locomotion?
- **CH:** 擁有較重的頭部會如何影響蛇的移動？
- **Answer (EN):** A heavy head creates greater anterior torque, which physically limits how far the snake can reach horizontally.
- **Answer (CH):** 較重的頭部會產生更大的前段扭矩，這在物理上限制了蛇能水平延伸的距離。

11. 例外發現：蟒蛇 (The Exception: Boa Constrictor)

- **EN:** Why is Snake D (*Boa constrictor*) considered a morphological exception in your study?
- **CH:** 為什麼蛇 D (蟒蛇) 在您的研究中被認為是形態上的例外？
- **Answer (EN):** It has a morphological disadvantage—a massive body and a short tail—yet it still performs exceptionally well in gap crossing.
- **Answer (CH):** 牠具有形態上的劣勢——龐大的身軀與短尾巴——但牠在穿越間隙時依然表現得非常出色。

12. 肌肉代償機制 (Muscular Compensation)

- **EN:** How does the *Boa constrictor* overcome its morphological disadvantages?
- **CH:** 蟒蛇是如何克服其形態劣勢的？
- **Answer (EN):** It overcomes them through "Muscular Compensation." It uses superior muscular tension to counteract the anterior torque.
- **Answer (CH):** 牠透過「肌肉代償」來克服。牠利用卓越的肌肉張力來對抗前段扭矩。

13. 核心結論 (Core Conclusion)

- **EN:** What is the ultimate conclusion regarding morphological trade-offs in snakes?
- **CH:** 關於蛇類的形態權衡 (Morphological trade-offs)，最終的結論是什麼？
- **Answer (EN):** A long tail provides a mechanical anchor, while a heavy body limits reach, unless the snake possesses extreme muscular tension to compensate.
- **Answer (CH):** 長尾巴提供了力學錨點，而沉重的身體會限制延伸範圍，除非蛇具備極端的肌肉張力來進行代償。

14. 實際應用價值 (Practical Application)

- **EN:** How can your biomechanical data be used in real life?
- **CH:** 您的生物力學數據可以如何應用於現實生活中？
- **Answer (EN):** We can use the 40% reach limit data to design scientifically-informed anti-snake barriers for human settlements.
- **Answer (CH):** 我們可以利用 40% 延伸極限的數據，為人類聚落設計具有科學依據的防蛇圍欄。

15. 聯合國永續發展目標 (SDG Alignment)

- **EN:** How does your project support the UN Sustainable Development Goals?
- **CH:** 您的專案如何支持聯合國永續發展目標？
- **Answer (EN):** It aligns with SDG 15: Life on Land, by promoting a "Scientifically Informed Coexistence" between humans and local wildlife.
- **Answer (CH):** 它符合 SDG 15：保育陸域生態，藉由提倡人類與當地野生動物之間「建立在科學基礎上的共存」。

💡 評審應對小建議 (Tips for Q&A)：

這張海報最精彩的科學思辨在於**「力學極限 (Mechanical Limit)」與「肌肉代償 (Muscular Compensation)」**。

當日本評審問到第 11 或 12 題（關於 Boa constrictor 蟒蛇）時，您可以面帶微笑、雙手做出一個「用力」的手勢並回答：*"According to our math, the Boa should fail because its tail is too short. But biology is amazing—the Boa simply uses brute muscular strength to break our morphological rules!"* 這種帶有幽默感且點出生物多樣性奧妙的回答，絕對能讓評審眼睛一亮！祝團隊赴日比賽大獲全勝！